

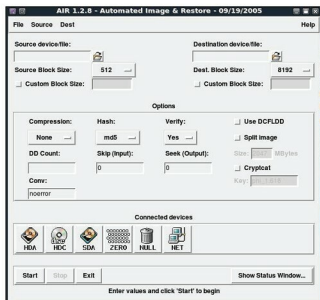


Figure 8: AIR Image GUI

are used to hold information about the disk image, and data segments called 'pages', which are used to hold the image disk information itself. Some of the default metadata segments that are created by the *aimage* utility and are included in the AFF image file are:

<i>acquisition_date</i>	Holds the date and time of imaging.
<i>Pagesize</i>	Holds the size of an uncompressed data segment, viz., a page (default 16 MB, same for all pages).
<i>imagesize</i>	size (in bytes) of the complete image (64-bit value).
<i>image_gid</i>	globally unique identifier that is different for every run of 'aimage'
<i>acquisition_device</i>	The device used as the source for imaging.
<i>Badsectors</i>	Total number of 512-byte bad sectors in the image (32-bit value).
<i>Badflag</i>	Pattern for bad sectors in the data segment(s).
<i>dir</i>	AFF directory segment.

AFF file data segments ('pages') can be compressed with the open source utility 'zlib', or can be left uncompressed. The segment flag of a data segment carries information about whether the data is compressed or not. Compressed AFF files consume less space, but are slower to create and access. The basic usage of 'aimage' is *aimage <options> <input> <outfile>*, where *<input>* is the input media (a drive/partition on a drive/data coming in on a TCP port) to be imaged; *<outfile>* is the output file; *<options>* is a list of options. Some of the important options are:

<i>--quiet, -q</i>	No interactive statistics.
<i>--no_dmesg, -D</i>	Do not put the 'dmesg' output into the AFF output file.
<i>--no_compress, -x</i>	Do not compress (useful for slow machines).
<i>--skip=<nn>[s], -k <nn>[s]</i>	Skip <i>nn</i> bytes from start of input media. Use the suffix 's' to specify sectors instead of bytes.
<i>-X<n></i>	Level of compression (<n> can range from 1 to 9).
<i>--error=0, -e0</i>	Standard error recovery.
<i>--error=1, -e1</i>	Stop reading at first error.
<i>-z</i>	Overwrite an already existing '.aff' output file.
<i>-s <name>=<value></i>	Create segment <name> with <i>value</i> . (The option can be repeated.)
<i>-L</i>	Use LZMA compression.
<i>-S <nn>, --image_page_size=<nn></i>	Specify the AFF file page size as <nn>, with optional suffix of <i>b</i> , <i>k</i> , <i>m</i> or <i>g</i> to reflect bytes, kilo-bytes, megabytes or gigabytes.

Example: To create the AFF image file 'image.aff' of the drive '/dev/sd0' with an additional metadata segment with the name 'case_number' and value 'abcd', invoke *aimage* as follows:

```
aimage /dev/sd0 image.aff -s case_number=abcd
```

Aimage also offers a real-time display to show the current status of imaging, as shown in Figure 5.

Afconvert

'Afconvert' allows converting of raw (bit-by-bit copy) forensic disk images (made using 'dd' -like tools or ISO images) into AFF format images, and vice-versa. By default, 'afconvert' uses gzip/bzip compression. However, LZMA is an alternative compression algorithm available, using the *-L* option, for better compression rates. Basic usage for 'Afconvert' is *afconvert <options> <input raw image>*, where *<input raw image>* is the raw image to be converted into AFF image file and *<options>* is a list of options. Some of the useful options are:

<i>--no_compress, -x</i>	Do not compress (useful for slow machines).
<i>-X<n></i>	Level of compression (<n> can range from 1 to 9).
<i>-z</i>	Overwrite an already existing '.aff' output file.
<i>-s <name>=<value></i>	Create segment <name> with <i>value</i> . (The option can be repeated.)